UNIT 2: BONDING

C. Lewis Structures
   1. Lewis structures and Formal Charge (7.3)
   2. Resonance (7.4)
   3. Exceptions to the Octet Rule (7.3)

D. Valence Shell Electron Pair Repulsion Theory (VSEPR) (7.6)
   1. Strategy
   2. Electron Group Arrangements
   3. Molecular Shape
   4. Bond Angles and Molecular Polarity
   5. Examples

E. Valence Bond Theory and Orbital Hybridization (8.1, 8.2)

Generate Lewis structures for covalent molecules and ions and, coupled with VSEPR and Valence Bond Theory, determine the molecule's geometry, polarity, and bonding (bond order, bond length, strength and energy).

C. Lewis Structures (Sec. 7.3, 7.4)

1) Writing Lewis Structures

- Only **valence electrons** are shown. For main-group elements, the number of valence electrons equals the "group number" or "group number minus 10." The nucleus and core electrons are represented by the symbol of the element, and the valence electrons are represented by dots.

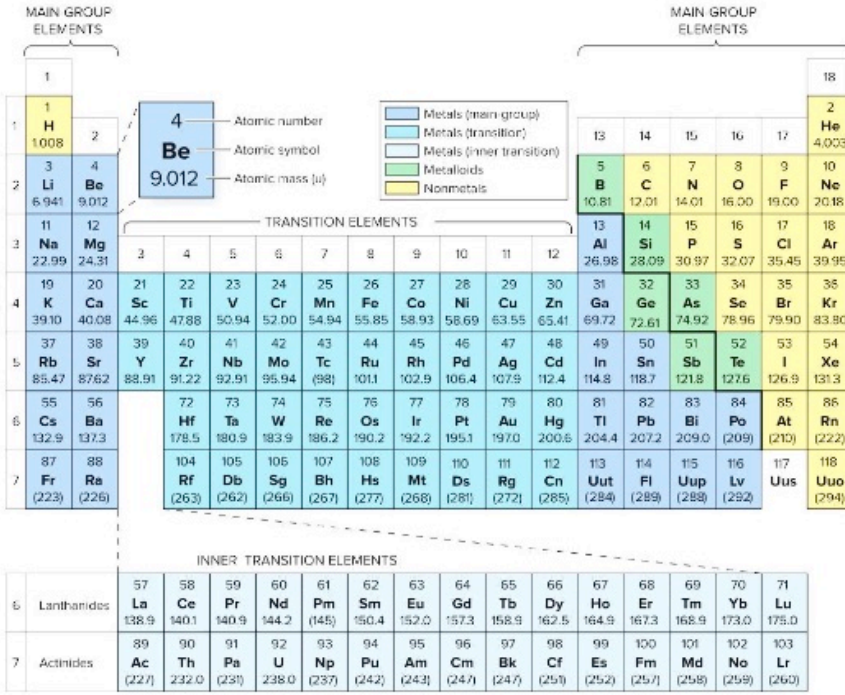
- A pair of **bonding electrons** (or a **bond pair**) between two atoms is a single covalent bond and is shown by a line. A pair of **non-bonding electrons** on one atom is a **lone pair** and is shown by two dots.

- Electrons are distributed so that atoms acquire a stable electron configuration, usually an **octet** (8 e^-) for most atoms (but 2 e^- for H atoms). Multiple bonds (double or triple bonds) may need to be formed.

Strategy:

1. Calculate total number of valence electrons. (Recall for main group elements, the number of valence electrons = group number or group number – 10). Add or subtract electrons if species is an anion or cation.
2. Identify central atom and write skeleton structure using SINGLE bonds.
   Carbon is usually a central atom
   Hydrogen is always a terminal atom
   If you have to choose, put the *least electronegative* atom as the central atom.
3. Determine electrons remaining. Total valence electrons – electrons needed for single bonds.
4. With remaining electrons, complete octets on TERMINAL atoms first. If any electrons are left over, complete the octet on the central atom.
5. If the central atom does not have an octet at this point, convert lone pairs on the terminal atoms into multiple bonds.

DATA SHEET



Equations

$E = h\nu$

$c = \lambda\nu$

$\lambda = \dfrac{h}{mu}$

$E_n = -R_H \dfrac{Z^2}{n^2}$ (where $R_H = 2.179 \times 10^{-18}$ J = 13.60 eV)

$x_i = \dfrac{n_i}{n_{tot}} = \dfrac{p_i}{p_{tot}}$

$u_{rms} = \sqrt{\dfrac{3RT}{M}}$

$pV = nRT$

$p = \dfrac{nRT}{V - nb} - \dfrac{an^2}{V^2}$

$(KE)_{avg} = \frac{3}{2}RT$

fc = nve − (unshared e⁻ + ½ shared e⁻)

bo = ½(bonding e⁻ − antibonding e⁻)

$\Delta H°_{rxn} \approx \Sigma BE(\text{broken}) - \Sigma BE(\text{formed})$

Average bond energies (kJ/mol)

Bond	Energy	Bond	Energy
C–C	347	C–O	358
C=C	614	C=O	745 (799 in CO_2)
C≡C	839	N–O	201
N–N	160	N=O	607
N=N	418	C–H	413
N≡N	945	N–H	391
O–O	204	O–H	467
O=O	498	H–H	432

Physical constants

$N_A = 6.022 \times 10^{23}$ mol^{-1}

$R = 0.08206$ L·atm·mol^{-1}·K^{-1}

$R = 8.314$ J·mol^{-1}·K^{-1}

$h = 6.626 \times 10^{-34}$ J·s

$c = 2.998 \times 10^{8}$ m·s^{-1}

$m_e = 9.109 \times 10^{-31}$ kg

$e = 1.602 \times 10^{-19}$ C

$m_p = 1.672 \times 10^{-27}$ kg

$m_n = 1.675 \times 10^{-27}$ kg

$a_o = 52.9$ pm = 0.529 Å

Conversion factors

0.0 °C = 273.15 K

1 nm = 10^{-9} m; 1 Å = 10^{-10} m = 10^{-8} cm ; 1 pm = 10^{-12} m

1 cal = 4.184 J

1 atm = 760 torr = 101.3 kPa

1 eV = 1.602×10^{-19} J = 96.5 kJ mol^{-1}

Visible spectrum

Wavelength (nm)	Colour
400	violet
450	blue
490	blue-green
570	yellow-green
580	yellow
600	orange
650	red

H_2CO	# valence e- # bonding e- electrons remaining:

Skeleton Structure	Completed Lewis Structure	Formal Charges

CN^-	# valence e- # bonding e- electrons remaining:

Skeleton Structure	Completed Lewis Structure	Formal Charges

NOCl	# valence e- # bonding e- electrons remaining:

Skeleton Structure	Completed Lewis Structure	Formal Charges

Formal Charge

Calculating Formal Charge:

1. Assign electrons to each atom.

Unshared electrons (lone pairs) are assigned to the atom on which they lie.

Bonding electrons get split between the atoms.

2. The difference between the number of valence electrons in the free atom and the number of assigned electrons is the formal charge.

Formal Charge of an atom =

(# of valence e-) – (# of unshared valence e- + ½ # of shared valence e-)

3. The sum of the formal charges for a neutral molecule =0

The sum of the formal charges for an ion = the charge on the ion.

**Calculate formal charges for all atoms in each example above.

Using Formal Charge:

Use formal charge to pick the best Lewis structure and/or assess the validity of your structure.

Keep formal charge as small as possible.

Keep negative formal charge on the more electronegative atom if possible.

2. Resonance (7.4)

Many times it is possible to draw more than one valid Lewis structure.

Example: NO_3^- (nitrate)

•It looks like NO_3^- has two single bonds and one double bond.

•Experimental evidence shows that NO_3^- has three equivalent N-O bond lengths.

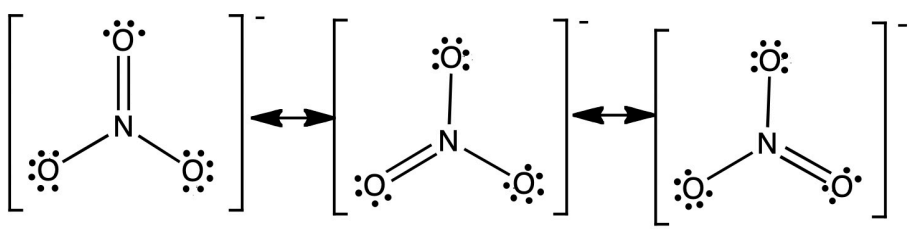
Bond Order = the number of electron pairs shared by two bonded atoms.

$$\text{Bond Order} = \frac{\text{number of shared e- pairs}}{\text{number of bonded atom pairs}}$$

Bond	Bond Order
C—O	1
C=O	2
C≡O	3

e.g. (triple bond) $\frac{3}{1} = 3$

What about the bond order in nitrate? It is a *fractional* bond order.



Example: NCO^-

3. Exceptions to the Octet Rule (7.3)

(a) Odd-electron Species

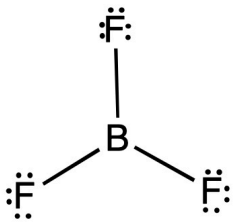
Odd electron Species—cannot satisfy the octet rule.

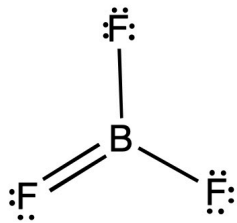
- Give as close to an octet as possible without going over.
- Odd electron compounds are extremely reactive.

Example: NO (nitric oxide)

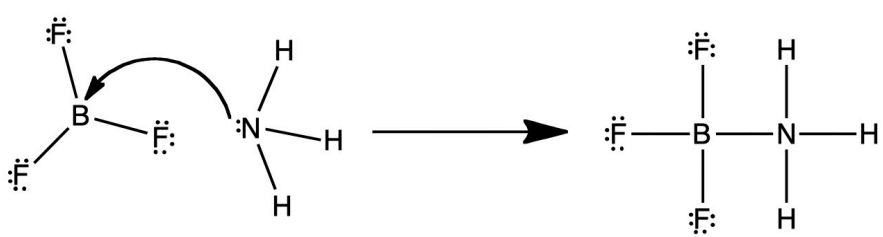
(b) Electron Deficient Compounds (B, Al, Be compounds)

example: BF_3





- Experimental evidence supports both structures. Although the second structure is unlikely (based on formal charges), it could be a **minor** contributor because of shorter than expected B-F bond observed in some data.
- BF_3 is very reactive toward **electron donors** suggesting that BF_3 is **electron deficient** (first structure).



Lewis acid-base reaction—one species donates an electron pair to another to form a covalent bond.

(c) Expanded octets—3rd row elements and higher can have expanded octets (>8 electrons).
2nd row elements cannot form expanded octets.

•Empty valence d-orbitals must be available for the surplus electrons.
•2nd row elements do not have allowed valence d-orbitals so cannot expand octets.

Example: PCl_5

•Sometimes you can use expanded octets to write alternative structures with minimal formal charge.

Example: SO_4^{2-}

D. Valence-Shell Electron-Pair Repulsion (VSEPR) Theory (Section 7.6)

- Molecular shape is the 3-dimensional arrangement of nuclei joined by bonding groups.
- Use VSEPR Theory to predict molecular shape.
- Basic principle: each group of valence electrons around a central atom is **located as far as possible** from the others to minimize repulsions.

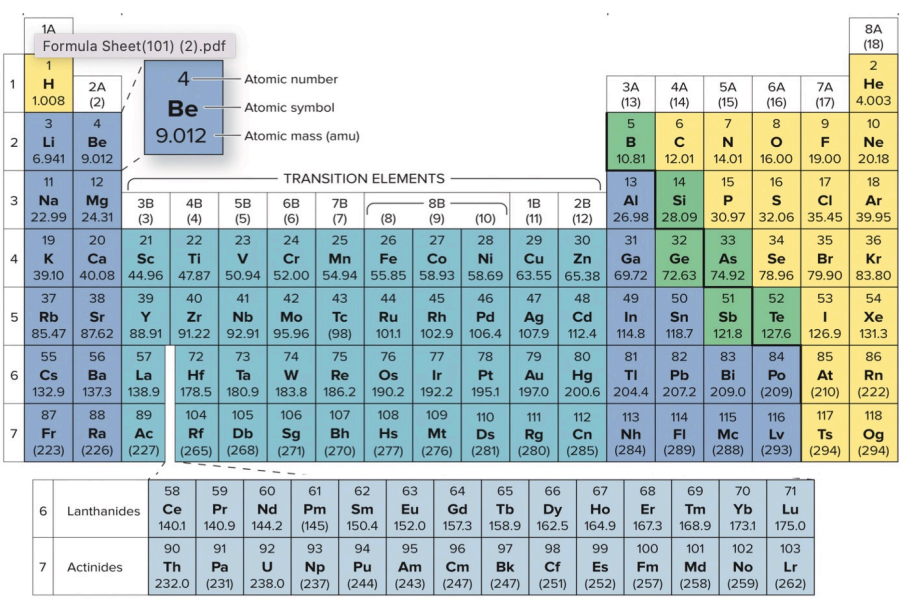
1. Strategy:

—Draw the Lewis Structure

—Count electron groups around a central atom: **a "group" of electrons is any number of electrons around a central atom**. Examples of **ONE electron group** are: single bond, double bond, triple bond, lone pair, single electron.

—Determine the electron group arrangement: each group of valence electrons around a central atom is located as far away as possible from the others in order to minimize repulsions. (see next slide)

—Determine molecular shape



2. Electron group arrangements

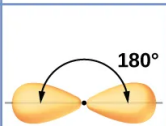
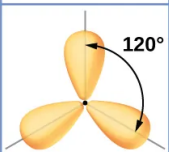
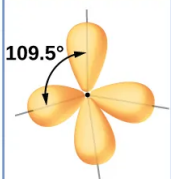
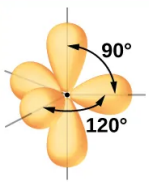
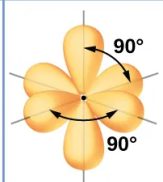
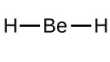
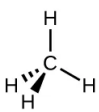
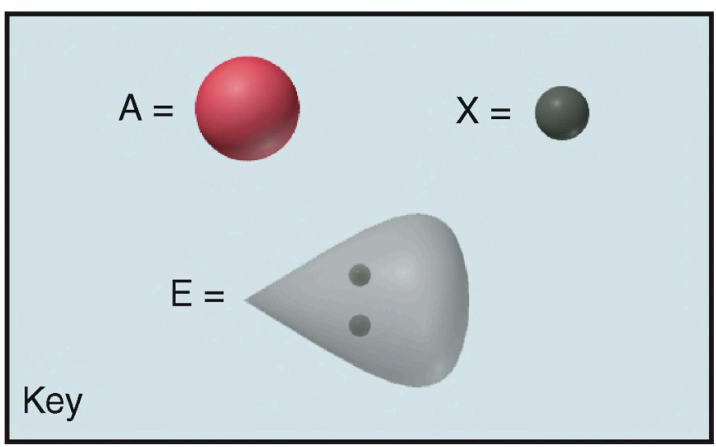
Number of regions	Two regions of high electron density (bonds and/or unshared pairs)	Three regions of high electron density (bonds and/or unshared pairs)	Four regions of high electron density (bonds and/or unshared pairs)	Five regions of high electron density (bonds and/or unshared pairs)	Six regions of high electron density (bonds and/or unshared pairs)
Spatial arrangement					
Line-dash-wedge notation					
Electron region geometry	Linear; 180° angle	Trigonal planar; all angles 120°	Tetrahedral; all angles 109.5°	Trigonal bipyramidal; angles of 90° or 120° An attached atom may be equatorial (in the plane of the triangle) or axial (above or below the plane of the triangle).	Octahedral; all angles 90° or 180°

Figure 7.16 The basic electron-pair geometries predicted by VSEPR theory maximize the space around any region of electron density (bonds or lone pairs).

3. Determine molecular shape: after determining electron group geometry, determine the molecular shape. Molecular shape: consider only those positions occupied by atoms.

AX_mE_n

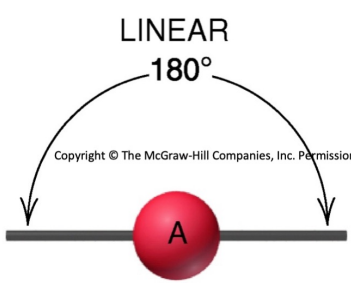
A = central atom
X = surrounding atom
E = nonbonding valence-electron group
m and *n* are integers



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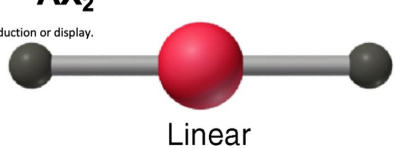
Electron Group Arrangement:

Linear



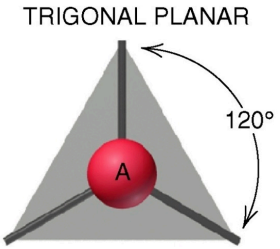
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AX_2

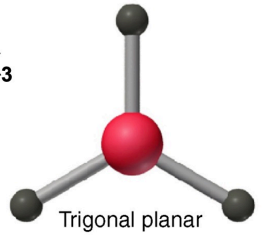


Electron Group Arrangement:

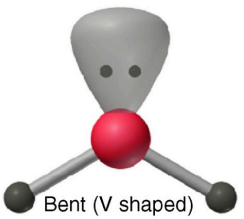
Trigonal Planar



AX_3

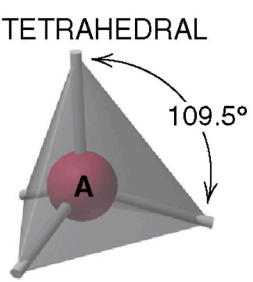


AX_2E

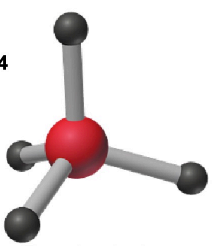


Electron Group Arrangement:

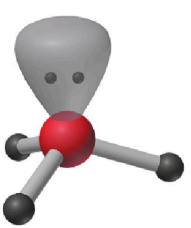
Tetrahedral



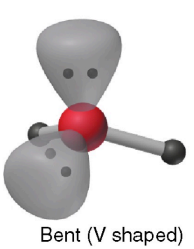
AX_4



AX_3E

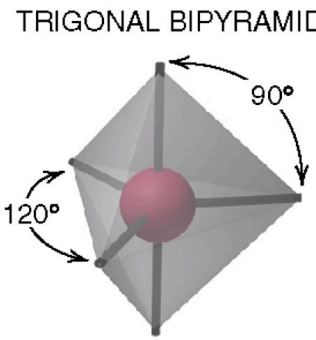


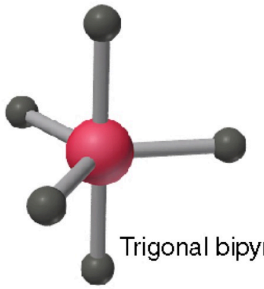
AX_2E_2



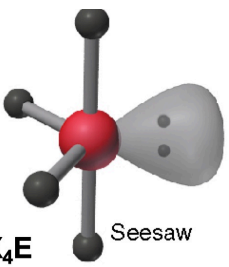
Electron Group Arrangement:

Trigonal bipyramidal

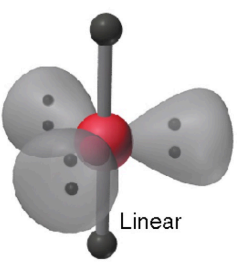




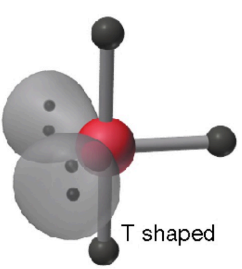
AX_5



AX_4E



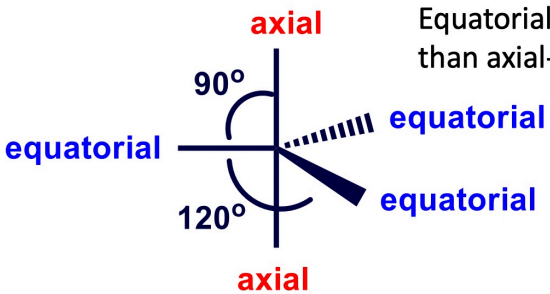
AX_2E_3



AX_3E_2

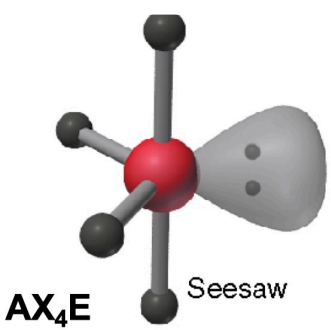
Axial and Equatorial Positions

A five electron-group system has two different positions for electron groups, and two ideal bond angles.

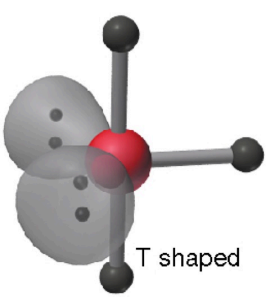


Equatorial-equatorial repulsions are weaker than axial-equatorial repulsions.

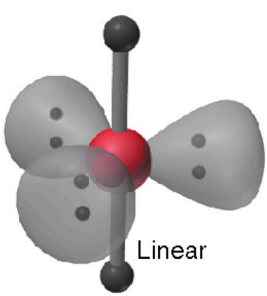
Where possible, lone pairs in a five electron-group system occupy *equatorial* positions.



AX_4E

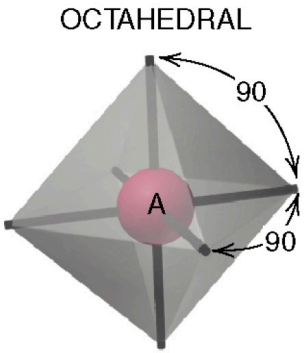


AX_3E_2

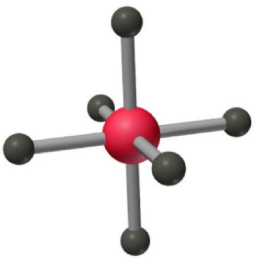


AX_2E_3

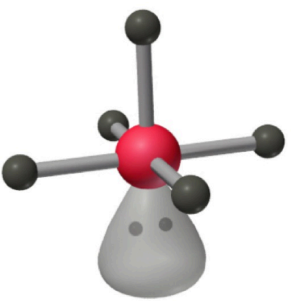
Electron Group Arrangement:
Octahedral



AX_6

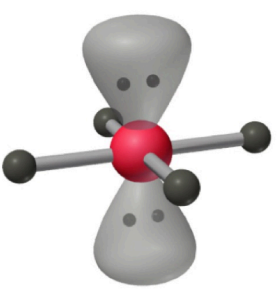


Octahedral



Square pyramidal

AX_5E



Square planar

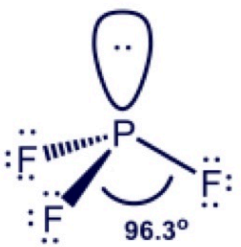
AX_4E_2

4. Bond Angles and Molecular Polarity:

The ***bond angle*** is the angle formed by the nuclei of two surrounding atoms with the nucleus of the central atom.

The angles given thus far are ***ideal*** bond angles, determined by basic geometry alone. Real bond angles deviate from the ideal value in many cases.

Nonbonding (Lone) Pairs

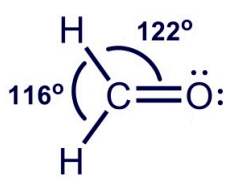


The ideal angle is 109.5°

AX_3E

A lone pair repels bonding pairs more strongly than bonding pairs repel each other. This decreases the angle between the bonding pairs.

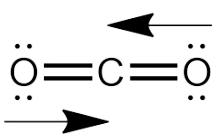
Double Bonds



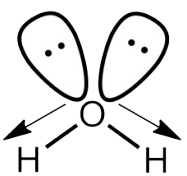
The ideal angle is 120°.

A double bond has greater electron density than a single bond, and repels the single bond electrons more than they repel each other.

- Overall ***molecular polarity*** depends on both ***shape*** and ***bond polarity***.
- The polarity of a molecule is measured by its ***dipole moment*** (**μ**), which is given in the unit *debye* (D).
- A molecule is polar if:
  - the individual bond dipoles ***do not cancel***.
- A polar molecule will orient itself in an electric field:
  (+) end of the molecule is attracted to (-) plate
  (-) end of the molecular is attracted to (+) plate

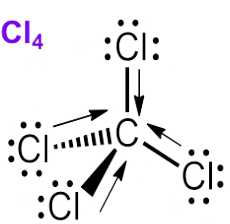


Individual bond polarities cancel. Therefore the molecule has ***no net dipole moment*** (μ = 0 D).



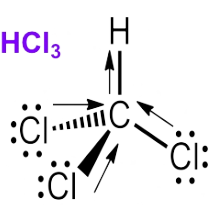
V shaped geometry: individual bond polarities do **not** cancel. ***Overall molecular polarity***. The O is partially negative while the H atoms are partially positive. (μ = 1.85 D).

CCl_4



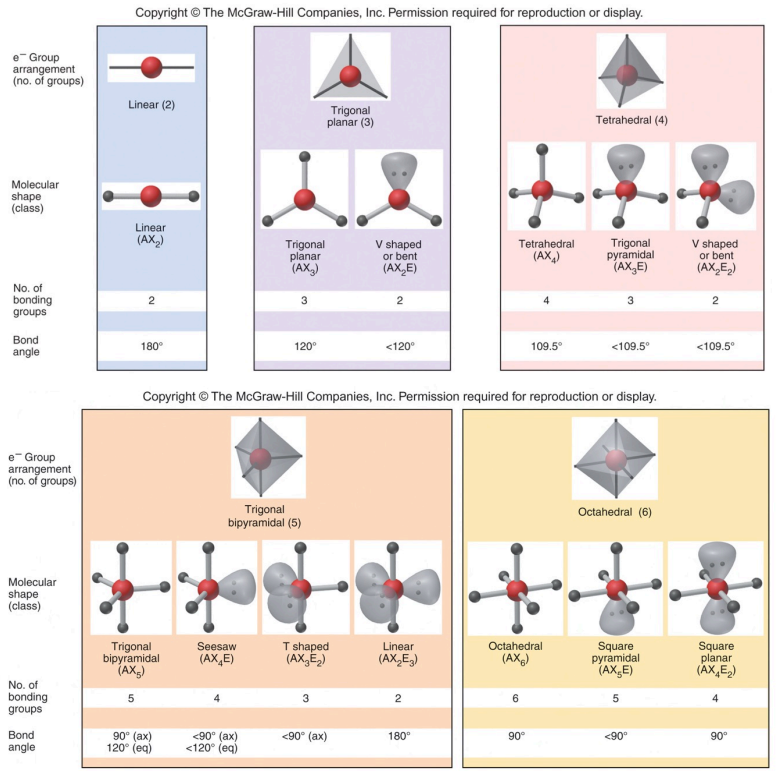
Bonds are polar, but individual bond polarities cancel.

$CHCl_3$



Bond polarities do not cancel. This molecule is polar (μ = 1.01 D).

Figure 9.9 **A summary of common molecular shapes with two to six electron groups.**



Silberberg McGraw Hill, 3rd Canadian Ed.

Write Lewis Structure and analyze the following:

Electron group arrangement

Molecular shape

AX_mE_n notation

Bond angles

Polar? If so, draw the dipole moment

5. Molecules with greater than one central atom.

Example: CH_3COOH

E. Valence Bond Theory and Orbital Hybridization (8.1, 8.2)

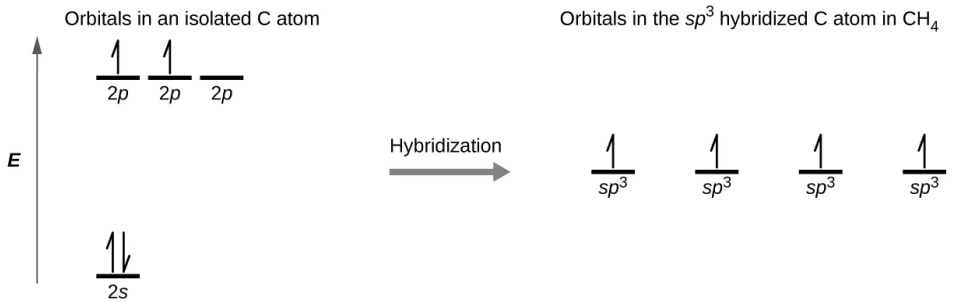
- Lewis structures and VSEPR allow us to predict geometry, but they do not tell us which atomic orbitals are involved in bonding.
- The basic principle for Valence Bond Theory is that a covalent bond forms when orbitals of two atoms overlap, and a pair of e^- is localized in the region between the atoms.
  (1) The pair of e^- has opposite spins.
  (2) Good, in-phase overlap leads to strong bonding interactions. (Phase is related to the sign of the wavefunctions of the atomic orbitals).
  (3) Orbitals can hybridize to more appropriate shapes and orientations, to maximize overlap.
  Direct overlap of the *s* and/or *p* orbitals of isolated atoms does not account for the shape of many molecules (e.g., **methane**)

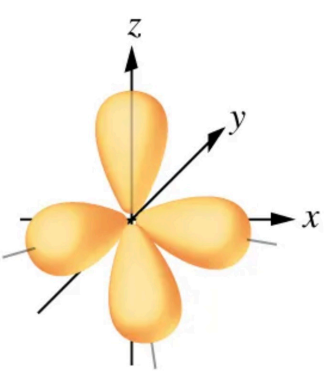
Carbon ([He]$2s^2 2p^2$) overlapping with H ($1s^1$) to form CH_4

- Linus Pauling proposed: during bonding, valence atomic orbitals (AOs) in the isolated atoms become *different* when they are in the molecule.

- Quantum-mechanical calculations show that if we mathematically "mix" certain combinations of orbitals, we form new orbitals whose spatial orientations match the observed molecular shapes.

- The process of orbital mixing is **called hybridization, and the new AOs are called hybrid orbitals.**





Hybridization is not an actual physical process; it is a hypothetical process that is invoked to explain molecular shapes that are observed experimentally.